

Conditional Statements

Introduction to Programming

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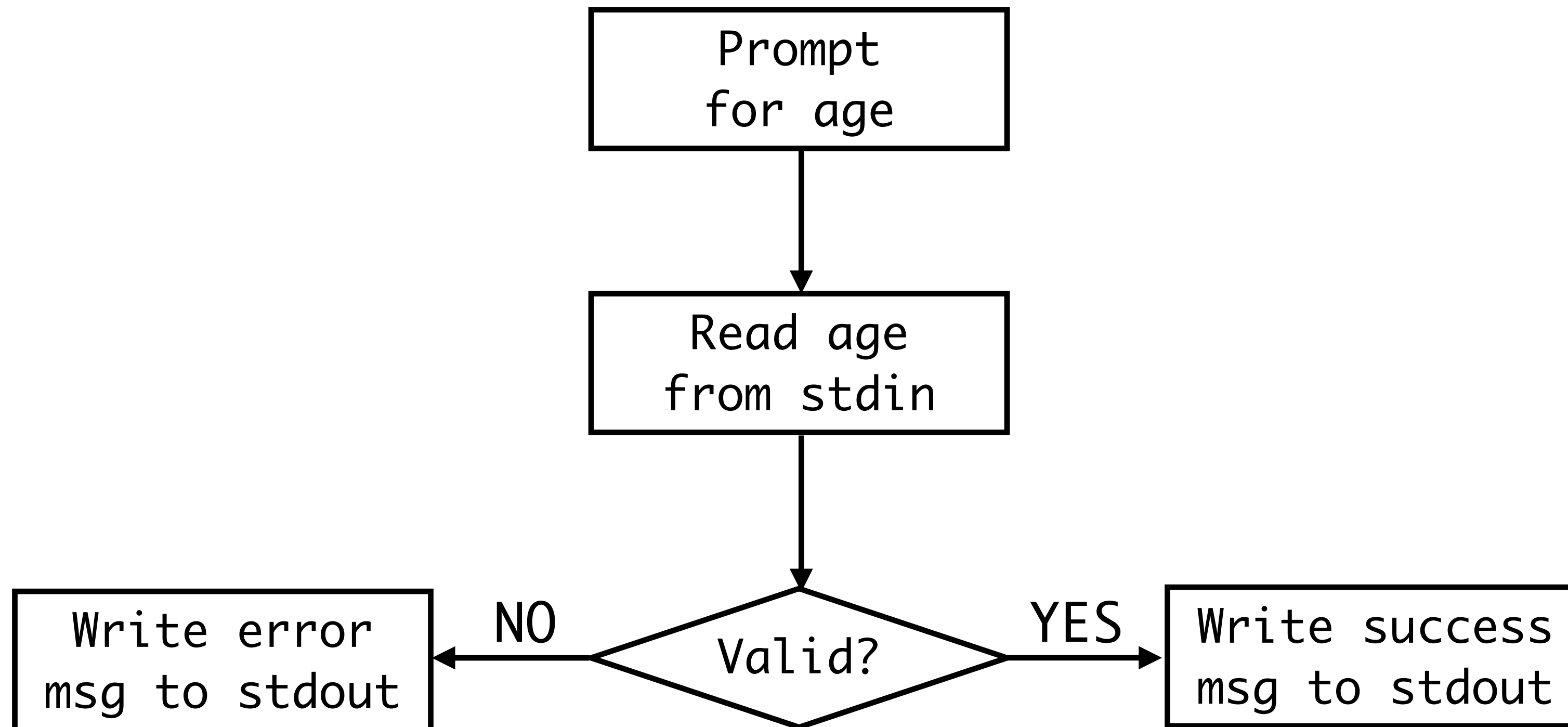
Introduction

Conditional Statements

- **Conditional statement:**
 - A statement comprised of n blocks
 - At most one of n blocks will be executed

Introduction

Conditional Statements



If Else

Conditional Statements

- Pseudocode, e.g.:
 - IF Temp > 20 THEN
 - OUTPUT "It's warm" to stdout
 - ELSE
 - OUTPUT "It's cool" to stdout
 - ENDIF

If Else

Conditional Statements

- PY: `if day_of_week == 6 or day_of_week == 7:`
- PY: `print("It's the weekend!")`
- PY: `else:`
- PY: `print("It's a weekday")`

If Else

Conditional Statements

- JS: `if (dayOfWeek == 6 || dayOfWeek == 7) {`
- JS: `console.log("It's the weekend!");`
- JS: `} else {`
- JS: `console.log("It's a weekday");`
- JS: `}`

If Else

Conditional Statements

- JV: `if (dayOfWeek == 6 || dayOfWeek == 7) {`
- JV: `System.out.println("It's the weekend!");`
- JV: `} else {`
- JV: `System.out.println("It's a weekday");`
- JV: `}`

If Else

Conditional Statements

- C#: `if (dayOfWeek == 6 || dayOfWeek == 7) {`
- C#: `Console.WriteLine("It's the weekend!");`
- C#: `} else {`
- C#: `Console.WriteLine("It's a weekday");`
- C#: `}`

Code Blocks

Conditional Statements

- PY: `if day_of_week == 6 or day_of_week == 7:`
- PY: `# true path code block`
- PY: `print("It's the weekend!")`
- PY: `else:`
- PY: `# false path code block`
- PY: `print("It's a weekday")`

One Path

Conditional Statements

- PY: `if day_of_week == 5:`
- PY: `print("It's nearly the weekend!")`

Three or More Paths

Conditional Statements

- PY: `if day_of_week == 5:`
- PY: `print("It's nearly the weekend!")`
- PY: `elif day_of_week == 6 or day_of_week == 7:`
- PY: `print("It's the weekend!")`
- PY: `else:`
- PY: `print("It's a weekday")`

Three or More Paths

Conditional Statements

- JS: `if (dayOfWeek == 5) {`
- JS: `console.log("It's nearly the weekend!");`
- JS: `} else if (dayOfWeek == 6 || dayOfWeek == 7) {`
- JS: `console.log("It's the weekend!");`
- JS: `} else {`
- JS: `console.log("It's a weekday");`
- JS: `}`

Three or More Paths

Conditional Statements

- JV: `if` (dayOfWeek == 5) {
- JV: `System.out.println("It's nearly the weekend!");`
- JV: } `else if` (dayOfWeek == 6 || dayOfWeek == 7) {
- JV: `System.out.println("It's the weekend!");`
- JV: } `else` {
- JV: `System.out.println("It's a weekday");`
- JV: }

Three or More Paths

Conditional Statements

- C#: `if (dayOfWeek == 5) {`
- C#: `Console.WriteLine("It's nearly the weekend!");`
- C#: `} else if (dayOfWeek == 6 || dayOfWeek == 7) {`
- C#: `Console.WriteLine("It's the weekend!");`
- C#: `} else {`
- C#: `Console.WriteLine("It's a weekday");`
- C#: `}`

Nesting

Conditional Statements

- PY: `if age < 18:`
- PY: `print("You are a minor")`
- PY: `else:`
- PY: `if age < 65:`
- PY: `print("You are of working age")`
- PY: `else:`
- PY: `print("You are eligible for retirement")`